Informational Power Through Pivotality: How Consensus Can Benefit a Hegemon

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Abstract

Why do hegemons build international organizations that decide by consensus, a rule that formally equalizes votes and grants them no agenda power? We study two-round bargaining in which only weak states propose and a hegemon holds private information about its outside option, comparing unanimity and majority rule under identical primitives. Under majority rule, proposers can buy uninformed weak-state votes instead of the hegemon's, and this substitute caps its price. Under unanimity, the hegemon is an essential input: the substitute disappears and its private threshold shapes the proposal. Relative to complete information, the institutional informational-rent contrast depends on the hegemon's type, the majority equilibrium class, and the parameter region. When the high type is sufficiently likely, it is positive for the low type when majority screens and can also be positive for the high type when public majority would exclude it; it vanishes when majority already pays the pooling price and may reverse under private exclusion. For an intermediate range of beliefs, unanimity has no perfect Bayesian equilibrium in pure ballot strategies, so the comparison is undefined there. Consensus can therefore convert pivotality into informational power, but only under conditions that the equilibrium correspondence makes explicit.

1 Introduction

Saudi Arabia is not formally entitled to write OPEC's production agreements, yet an agreement that ignored its spare capacity and outside opportunities would be difficult to sustain (Fattouh and Mahadeva 2013; Nakov and Nuño 2013). The World Trade Organization presents the mirror image: the United States accepted a consensus rule that formally gives every member an equal vote (Steinberg 2002). These cases sharpen a three-part puzzle. Why would a hegemon accept no agenda control, why accept a rule stricter than majority, and why insist on consensus[1] when that rule appears to equalize formal power?

The complete-information benchmark isolates the familiar answer. When the hegemon's strength is publicly known, unanimity makes its approval necessary and therefore adds veto power. Majority can instead leave the hegemon outside the agreement whenever buying weak-state votes is cheaper. The informational puzzle begins after this benchmark: what additional advantage arises because the other states do not know the hegemon's reservation value?

We study a two-round bargaining game with one hegemon and at least three weak states. Only weak states make proposals. The hegemon privately knows whether its terminal disagreement payoff is low or high. Proposals and simultaneous ballots are public, the institutional surplus is fixed, and a failed first round leads to a discounted terminal round. Majority and unanimity change only the number of votes required.

The mechanism is the difference between a substitute and an essential input. Under majority, a proposer can buy enough uninformed weak-state votes to pass without the hegemon. That outside coalition caps what the hegemon can command. Under unanimity, no such substitute exists: every agreement requires the hegemon's vote, so the proposal must respond to a privately known threshold. The resulting rent is frequently attached to pooling rather than to separation. A hegemon whose strength is overestimated receives the concession needed by the high type even when its own type is low.

The exact result is conditional. Majority has pure equilibria featuring exclusion, type-contingent delay, or pooling, depending on outside options and beliefs. Unanimity has an immediate low-type

[1] For the model, *consensus* and *unanimity* denote the same voting rule: every state must approve. The distinction between the terms does not affect the analysis.

agreement when the low type is certain, an immediate pooling agreement when the high type is sufficiently likely, and no perfect Bayesian equilibrium in pure ballot strategies in between. Against the public benchmark, unanimity's informational-rent advantage is therefore type- and region-specific. When the high type is sufficiently likely, it is positive for the low type when majority screens and also benefits the high type when public majority would exclude it, is zero when majority already pools, and can have either sign when private majority excludes. Where private unanimity has no pure-strategy equilibrium, the institutional contrast is empty rather than assigned a counterfactual payoff.

Our analysis builds directly on work showing how voting rules organize legislative bargaining and information aggregation (Baron and Ferejohn 1989; Winter 1996; Feddersen and Pesendorfer 1998; Eraslan and Evdokimov 2019). Miller, Montero, and Vanberg (2018) provide a useful complete-information benchmark for bargaining under heterogeneous voting rules. Closest to our informational setting, Piazolo and Vanberg (2025) analyze private information, rejection, and refinements in legislative bargaining, while Glynia, Thum, and Xefteris (2026) study the choice between insuring unanimity and gambling on a majority coalition in a one-shot setting. Their mechanisms and results receive full priority as the nearest points of comparison.

Our narrower contribution is to isolate a different margin: a privately informed actor that never proposes can gain because a voting rule either preserves or removes uninformed substitutes for its vote. Comparing each private-information game with its own complete-information counterpart then separates the familiar power of a necessary vote from the additional informational component. This distinction complements rational-design and informal-power accounts of international institutions (Koremenos, Lipson, and Snidal 2001; Steinberg 2002; Stone 2011).

The paper proceeds as follows. Section 3 defines the game and solution concept. Section 4 first solves the four public benchmarks, then the private games, and finally informational rents and their difference across voting rules. Section 5 interprets the essential-input mechanism and its limits. The appendix gives the proofs, endpoint discipline, exact correspondence tables, and a worked numerical illustration.

2 A working numerical illustration

Consider one hegemon and four weak states. The unit pie is bargained over with discount factor 0.9. The two disagreement payoffs are 0.10 and 0.35. In the terminal unanimity round, a proposer chooses between an offer accepted only by the low type and an offer accepted by both types. The cutoff probability of the high type is

$$\nu^* = \frac{0.35 - 0.10}{1 - 0.10} = 0.2778.$$

At a belief of 0.20, the proposer offers 0.10 and accepts the risk of failure when the type is high. At a belief of 0.35, it offers 0.35 and both types accept. This is a working numerical illustration, not an empirical calibration. The full two-round game adds the option to buy weak-state votes and the continuation created by a failed first ballot.

3 The model

3.1 Players, information, and proposals

There is one hegemon, H, and $m \geq 3$ weak states, $W = \{1, \dots, m\}$. Thus $N = m + 1$. Nature selects $\theta \in \{0, 1\}$, observed only by H, with $\Pr(\theta = 1) = \nu \in [0, 1]$. The terminal disagreement payoffs satisfy

$$0 < o_0 < o_1 < 1.$$

The game has two rounds. In each round, one weak state is recognized uniformly to propose; H is never a proposer. The two recognition draws are independent and made with replacement. Every weak state remains eligible in Round 2, including the Round-1 proposer. A proposal by weak state i is

$$s = (y, (x_j)_{j \in W \setminus \{i\}}, r_i), \qquad 0 \leq y \leq \bar{y}, \quad x_j \geq 0, \quad r_i \geq 0, \quad y + \sum_{j \neq i} x_j + r_i \leq 1,$$

where $o_1 \leq \bar{y} \leq 1$. Payments cannot be negative and side payments outside the proposed package are unavailable. Here y is the institutional concession assigned to H, x_j is the share assigned to responding weak state j, and r_i is the proposer's residual. The fixed unit pie is exhausted on the equilibrium path.

The payoff from agreement for H is $y + b_\theta$. We normalize $b_\theta = 0$, so an included hegemon receives the concession y. [AUTHOR: P2] This normalization isolates informational concessions and pivotality from any intrinsic benefit that an agreement might itself provide.

3.2 Ballots, timing, and payoffs

The proposer is counted as voting yes. All other states vote simultaneously; the complete vote vector and outcome become public only after all votes have been cast. Majority requires

$$q = \left\lfloor \frac{N}{2} \right\rfloor + 1$$

yes votes, while unanimity requires every state's yes vote. The ballot protocol is otherwise identical across rules.

If a proposal passes, every stated weak-state allocation is implemented. If H votes yes, it receives y. If H votes no but a majority passes the proposal without it, its general payoff is $y + o_\theta$. This formula is part of the complete game because the proposed institutional term y is implemented even without H's approval. In every equilibrium exclusion derived below, however, the proposer chooses $y = 0$; the hegemon therefore receives exactly o_θ, never a positive concession on top of it.

If nothing passes in Round 1, no payoff is realized then and the game proceeds to Round 2. Round-2 payoffs are multiplied by $\beta \in (0, 1)$ when evaluated in Round-1 units. If nothing passes in Round 2, each weak state receives zero and H receives o_θ in Round-2 units. A no vote is only a ballot action; it does not remove a player or trigger disagreement by itself. [AUTHOR: P1] The delayed terminal disagreement payoff represents the cost of prolonging international negotiations.

Table 1: Transitions and payoffs in the essential-input game

Event	Weak-state payoffs	H votes yes	H votes no
Proposal passes	Proposed x_j and r_i	y	$y + o_\theta$
Round-1 proposal fails	No current payoff; proceed to Round 2	$\beta C_H(h^Y)$	$\beta C_H(h^N)$
Round-2 proposal fails	Zero	o_θ	o_θ

Here h^Y and h^N denote the distinct public histories in which H voted yes or no, and $C_H(h^a)$ denotes H's Round-2 continuation value after history h^a. These continuation values need not be equal because the public vote can change beliefs before Round 2.

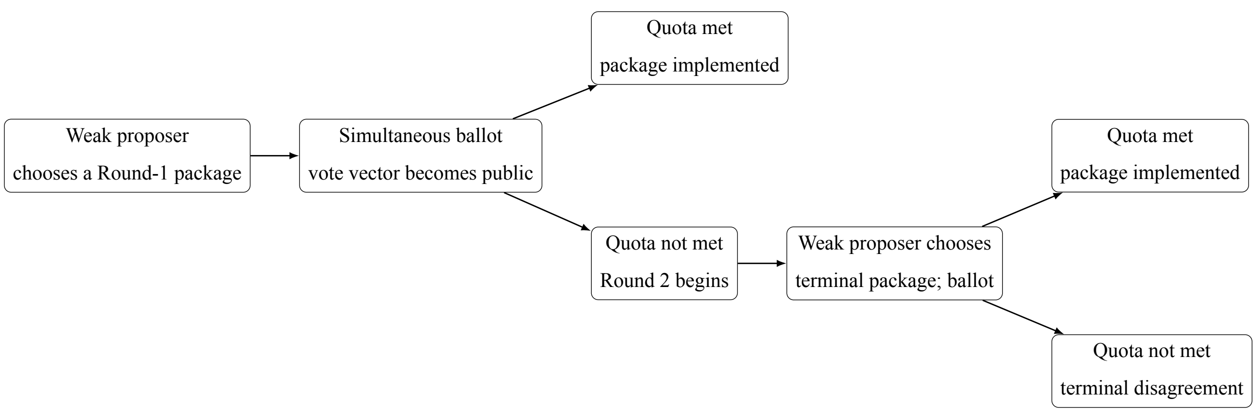


Figure 1: Sequence of proposals and ballots. Within each ballot, responding states vote simultaneously. A failed first-round proposal continues to the terminal round; a no vote alone does not end the game.

3.3 Solution concept

We use perfect Bayesian equilibrium with pure ballot strategies and three declared disciplines. First, an action by a player who does not know θ does not change beliefs about θ. After an action by H, beliefs follow Bayes' rule whenever possible and otherwise satisfy structural consistency. Second, a weak responder evaluates yes and no as if its own vote were pivotal. Third, exact expected indifference is resolved in favor of yes, denoted T^Y. Among proposal choices that give the proposer the same expected payoff, the selected proposal minimizes H's expected payoff.

Endpoint beliefs preserve the support of the prior. If $\nu = 0$, the posterior is zero at every history; if $\nu = 1$, it is one at every history. For $0 < \nu < 1$, an action by H that has zero probability under both types may support any posterior in $[0, 1]$. This is a condition of our solution concept. It follows the no-signaling logic used in multistage games (Fudenberg and Tirole 1991); support preservation is analogous to, but narrower than, the never-dissuaded discipline discussed by Osborne and Rubinstein (1990). Tremble consistency motivates the restriction but is not invoked as a theorem for degenerate priors (Kreps and Wilson 1982).

Table 2: Scope of the formal results

Element	Maintained scope
Players and horizon	One hegemon, $m \geq 3$ weak states, two rounds
Agenda	Only weak states propose in both rounds
Voting	Simultaneous public ballots; majority or unanimity
Information	Binary private type of H; support-preserving endpoints
Payoffs	Unit pie, $0 < o_0 < o_1 < 1$, $0 < \beta < 1$, $b_\theta = 0$
Equilibrium	Perfect Bayesian equilibrium in pure ballot strategies with the declared voting and proposal tie-break rules

4 Results

The solution proceeds by backward induction. We begin with public types because that benchmark separates the value of a necessary vote from the value of private information. We then solve the private terminal games, the private first-round games, and finally subtract the public benchmark component by component.

4.1 Complete-information benchmark

With a public type, no belief or signaling issue remains. The proposer compares the price of H's vote with the price of enough weak-state votes. In the terminal round, the relevant price of H is o_θ. In the first round, it is βo_θ, because rejection leads to the terminal round. Under majority, a weak-state vote costs β/m in first-round units. Under unanimity with public disagreement payoff o, each weak responder's continuation price is $\beta(1-o)/m$.

Proposition 4.1 (Public-type benchmark)**.** *Fix a public type with disagreement payoff o. In the terminal round, majority passes a proposal without H, assigns $y = 0$, and gives the weak proposer the whole unit pie; H receives o. Terminal unanimity passes with $y = o$, and the weak proposer retains $1 - o$.*

In Round 1, unanimity passes immediately with $y = \beta o$. Majority includes H if $o \leq 1/m$ and excludes it if $o > 1/m$. At equality the proposal including H is selected. Consequently, the public Round-1 payoff of H is

$$p_M(o) = \begin{cases} \beta o, & o \leq 1/m, \\ o, & o > 1/m, \end{cases} \qquad p_U(o) = \beta o.$$

Table 3: Complete-information outcomes by round and voting rule

Round	Rule	Proposed payments	Weak proposer	H
2	Majority	$y = 0$; weak responders get zero	1	o
2	Unanimity	$y = o$; weak responders get zero	$1 - o$	o
1	Majority, $o \leq 1/m$	$y = \beta o$; $q-2$ responders get β/m	$1 - \beta o - (q-2)\beta/m$	βo
1	Majority, $o > 1/m$	$y = 0$; $q-1$ responders get β/m	$1 - (q-1)\beta/m$	o
1	Unanimity	$y = \beta o$; every responder gets $\beta(1-o)/m$	$1 - \beta o - (m-1)\beta(1-o)/m$	βo

At $o = 1/m$, buying H's vote and buying the substitute weak-state vote cost the proposer the same amount. The proposal tie-break selects inclusion because it gives H the lower payoff $\beta o < o$.

4.2 Private information in the terminal round

Let

$$\nu^* = \frac{o_1 - o_0}{1 - o_0}.$$

Under terminal majority, H's vote is unnecessary and the belief never enters the proposer's choice. Under terminal unanimity, the proposer compares a low offer that succeeds only when the type is low with a high offer that both types accept.

Proposition 4.2 (Private terminal games)**.** *Under majority, the unique equilibrium outcome assigns $y = 0$, pays no responding weak state, and gives the full pie to the proposer. The proposal passes without H; type θ receives o_θ, and a weak state receives $1/m$ before recognition.*

Under unanimity, if $\nu \leq \nu^*$, the proposer offers $y = o_0$. The low type accepts and the high type rejects, so the proposal succeeds with probability $1 - \nu$. If $\nu > \nu^*$, the proposer offers $y = o_1$ and both types accept. At $\nu = \nu^*$, the low offer is selected.

The cutoff equates $(1-\nu)(1-o_0)$, the proposer's payoff from the low offer, with $1-o_1$, its payoff from pooling. Notice that no discount factor appears inside this terminal comparison.

4.3 Private majority in Round 1

Define

$$w = \frac{\beta}{m}, \quad t_\theta = \beta o_\theta,$$

and the proposer's payoffs from the three undominated outcome classes:

$$\begin{aligned} E &= 1 - (q-1)w, \\ L &= 1 - (q-2)w - t_0, \\ S(\nu) &= (1-\nu)L + \nu w, \\ P &= 1 - (q-2)w - t_1. \end{aligned}$$

Exclusion buys $q-1$ weak responders at price w each and sets $y = 0$. Screening buys $q-2$ weak responders and offers t_0: the low type accepts and the high type delays. Pooling buys $q-2$ weak responders and offers t_1, which both types accept. Deliberate delay is strictly worse than exclusion because $1 - \beta q/m > 0$.

When $o_0 < 1/m$, define

$$\nu_{SE} = \frac{\beta(1/m - o_0)}{\beta(1/m - o_0) + 1 - \beta q/m}.$$

When $o_1 < 1/m$, define

$$\nu_{SP} = \frac{\beta(o_1 - o_0)}{1 - \beta o_0 - \beta(q-1)/m}.$$

Proposition 4.3 (Private majority correspondence)**.** *For $m \geq 3$, the equilibrium outcome class is:*

1. *If $o_1 < 1/m$, screening for $\nu \leq \nu_{SP}$ and pooling above it.*
2. *If $o_0 < 1/m < o_1$, screening for $\nu \leq \nu_{SE}$ and exclusion above it.*
3. *If $1/m < o_0 < o_1$, exclusion for every ν.*
4. *If $o_0 = 1/m < o_1$, screening at $\nu = 0$ and exclusion for $\nu > 0$.*
5. *If $o_0 < o_1 = 1/m$, screening for $\nu \leq \nu_{SE}$. Above that cutoff, exclusion is selected when $(1-\nu)o_0 + \nu o_1 < \beta o_1$, pooling is selected when the inequality is reversed, and the entire proposal segment joining the two survives when the two expressions are equal.*

At every cutoff involving screening, screening is selected. Permuting the identities of identically paid weak responders generates additional equilibria but does not change H's payoff vector or the outcome class. Payoffs of weak states identified by label can be permuted and, on the residual proposal family, vary with the common proposal weight.

The last segment is genuine multiplicity. Its mixing weight is a probability over pure proposals in the proposer's strategy, and the same weight must be used jointly for payoffs and outcomes. It is not a license to combine componentwise minima and maxima into an unattained rectangle.

4.4 Private unanimity in Round 1

Let

$$\ell = \beta o_0, \qquad h = \beta o_1, \qquad A = \frac{\beta(1-o_0)}{m}, \qquad B = \frac{\beta(1-o_1)}{m}.$$

At $\nu = 0$, support preservation prevents an impossible high type from being reintroduced by an off-path belief. The proposer can therefore offer the low threshold ℓ, pay each responding weak state its continuation A, and keep $A+1-\beta$. When $\nu > \nu^*$, the continuation itself pools, so the proposer offers h, pays each responder B, and keeps $B+1-\beta$.

The intermediate cell fails for a different reason. A feasible deviation can pay every weak responder enough to force a yes vote. Once all weak votes are yes, no pure contingent voting profile for the two possible types of H survives: pooling yes is rejected by the high type, pooling no is overturned by the low type at equality, and either separating profile gives one type a profitable imitation.

Proposition 4.4 (Private unanimity correspondence)**.** *Under unanimity:*

$$\begin{aligned} \nu = 0 \quad &: \quad \textit{immediate low-type agreement with } y=\ell,\ x_j = A,\ r_i = A+1-\beta; \\ 0 < \nu \le \nu^* \quad &: \quad \textit{no perfect Bayesian equilibrium in pure ballot strategies}; \\ \nu^* < \nu \le 1 \quad &: \quad \textit{immediate pooling agreement with } y=h,\ x_j = B,\ r_i = B+1-\beta. \end{aligned}$$

The corresponding payoff vectors for H, ordered as low and high type, are (ℓ, h) at $\nu = 0$ and (h, h) above ν^*.

Remark (Pure-strategy scope). The middle cell establishes nonexistence only within the paper's maintained space of pure ballot strategies. We do not derive, characterize, select, or compare equilibria involving mixed ballot strategies, and make no claim about existence under such strategies.

Table 4: Private-information Round-1 equilibrium correspondences

Rule	Parameter cell	Selected outcome class	H payoff vector
Majority	$o_1 < 1/m$	Screening for $\nu \le \nu_{SP}$; pooling above	(t_0, t_1) or (t_1, t_1)
Majority	$o_0 < 1/m < o_1$	Screening for $\nu \le \nu_{SE}$; exclusion above	(t_0, t_1) or (o_0, o_1)
Majority	$1/m < o_0$	Exclusion for every ν	(o_0, o_1)
Majority	$o_0 = 1/m < o_1$	Screening at $\nu = 0$; exclusion for $\nu > 0$	(t_0, t_1) or (o_0, o_1)
Majority	$o_0 < o_1 = 1/m$	Screening through ν_{SE}; above, exclusion, pooling, or their exact proposal segment under the tie-break	(t_0, t_1), (o_0, o_1), (t_1, t_1), or the linked segment
Unanimity	$\nu = 0$	Immediate low-type agreement	(ℓ, h)
Unanimity	$0 < \nu \le \nu^*$	No PBE in pure ballot strategies	$\emptyset$
Unanimity	$\nu^* < \nu \le 1$	Immediate pooling agreement	(h, h)

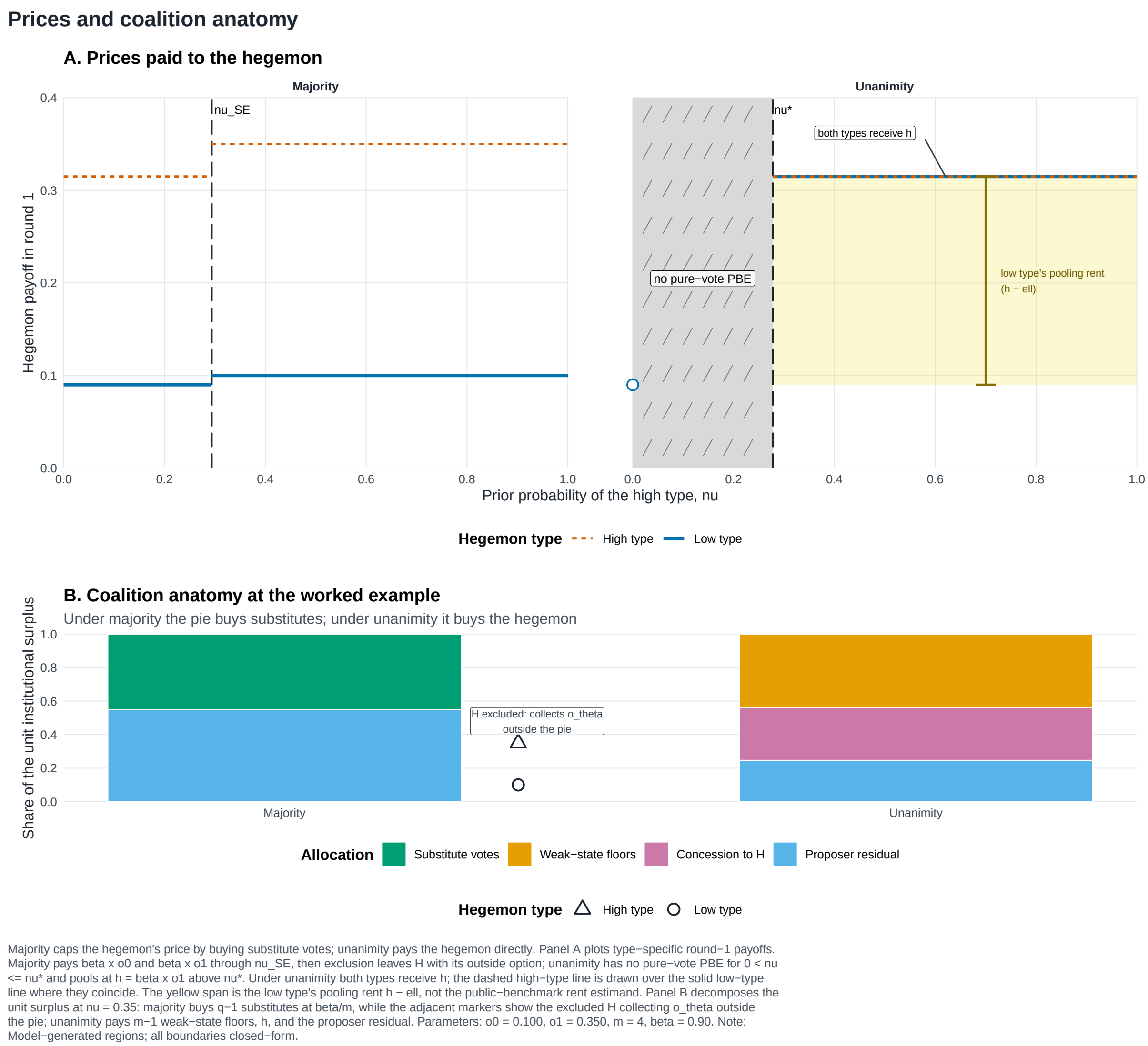


Figure 2: The price and coalition anatomy of the private games. Majority can replace the hegemon's vote with an additional weak-state vote; unanimity cannot. Labels report first-round prices and preserve the distinct low-type, pooling, and exclusion outcomes.

4.5 Comparing the private games

The comparison below uses the same parameter point under both institutions and only cells in which both games have a pure-strategy equilibrium. Let

$$a_\theta = (1-\beta)o_\theta, \qquad d = \beta(o_1 - o_0), \qquad k = \beta o_1 - o_0.$$

The terms a_θ are timing wedges: exclusion realizes the disagreement payoff now, whereas inclusion can price a continuation that is discounted in Round-1 units. They are distinct from the type gap d and the cross-date quantity k.

Proposition 4.5 (Private institutional payoff contrast)**.** *Write $V_U^{priv} - V_M^{priv}$ as a low-type/high-type vector.*

1. *At $\nu = 0$, the vector is $(0, 0)$ if majority screens and $(-a_0, -a_1)$ if majority excludes.*
2. *For $0 < \nu \leq \nu^*$, the contrast is empty because private unanimity has no pure-strategy equilibrium.*
3. *For $\nu > \nu^*$, the vector is $(d, 0)$ if majority screens, $(0, 0)$ if majority pools, and $(k, -a_1)$ if majority excludes.*
4. *On the residual exclusion–pooling segment at $o_1 = 1/m$, the exact attainable set is $\{\lambda(k, -a_1) : \lambda \in [0, 1]\}$, restricted to the proposal weights that survive the tie-break. The same λ binds payoffs and outcomes.*

Thus unanimity does not have a single unconditional payoff effect. Its effect depends on which substitute coalition disciplines the majority proposer and on which type is evaluated. Figure 3 maps these exact classes without averaging types.

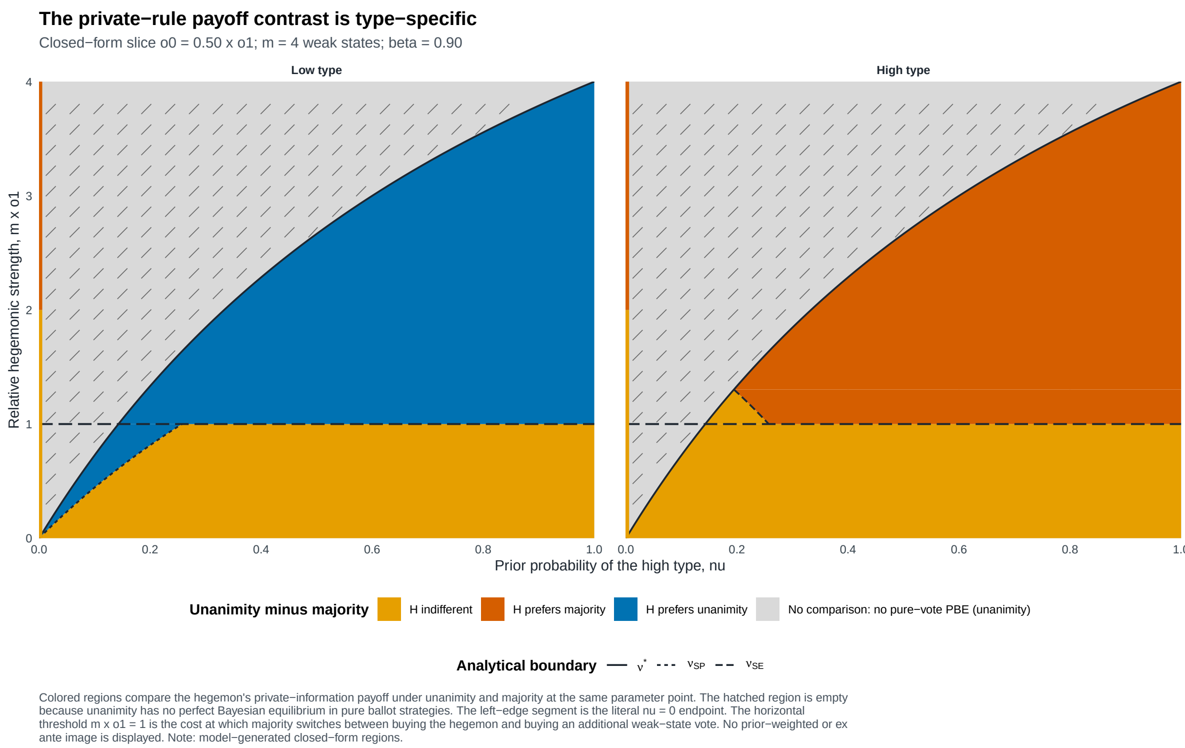


Figure 3: Private-game payoff contrast, unanimity minus majority, by hegemon type. Each cell is evaluated at the same parameter point. The hatched band marks parameter values for which private unanimity has no pure-strategy equilibrium and the contrast is therefore empty.

4.6 Informational rents and the difference of differences

For rule $g \in \{M, U\}$, define the type-contingent informational-rent correspondence

$$RI_g = V_g^{priv} - V_g^{pub},$$

where the public benchmark is evaluated at the same type and the same remaining parameters. The institutional informational-rent contrast is

$$\Delta RI = RI_U - RI_M.$$

These are differences between correspondences. If a source correspondence is empty, so is the difference. Under multiplicity, subtraction preserves the entire attainable set and its common proposal weight.

For compactness, call the public outside-option regions

$$\text{II} : o_1 \leq 1/m, \qquad \text{IX} : o_0 \leq 1/m < o_1, \qquad \text{XX} : 1/m < o_0.$$

The labels only record whether each public type is included by majority.

The rent comparison first asks what private information changes *within* each rule. Under unanimity, the public benchmark already buys the known type at its discounted threshold; private pooling therefore raises only the low type's payoff. Under majority, information matters only when the private equilibrium changes the public inclusion decision or the price paid to a type.

Proposition 4.6 (Informational rents by voting rule)**.** *Under unanimity,*

$$RI_U = \begin{cases} \{(0,0)\}, & \nu = 0, \\ \emptyset, & 0 < \nu \leq \nu^*, \\ \{(d,0)\}, & \nu > \nu^*. \end{cases}$$

Under majority, the informational rent depends jointly on the public region and the private equilibrium class:

$$\begin{aligned} \text{II, } \textit{screening} &: (0,0), \\ \text{II, } \textit{pooling} &: (d,0), \\ \text{II, } \textit{exclusion} &: (a_0,a_1), \\ \text{IX, } \textit{screening} &: (0,-a_1), \\ \text{IX, } \textit{exclusion} &: (a_0,0), \\ \text{XX, } \textit{exclusion} &: (0,0). \end{aligned}$$

On the residual exclusion–pooling segment at $o_1 = 1/m$*, the exact set is* $\{\lambda(a_0,a_1) + (1-\lambda)(d,0) : \lambda \in [0,1]\}$*, restricted to the segment that survives the proposal tie-break.*

The difference of differences then removes majority's informational rent from unanimity's. Screening can leave a positive low-type contrast in the high-belief cell because unanimity pools. Exclusion instead compares the high threshold discounted to the present with the low current disagreement payoff, so its low-type sign is governed by $k = \beta o_1 - o_0$.

Proposition 4.7 (Institutional informational-rent contrast)**.** *At $\nu = 0$, $\Delta RI = (0, 0)$ in regions II and XX, while $\Delta RI = (0, a_1)$ in region IX; the high-type component there is off support but remains part of the type-contingent vector.*

For $0 < \nu \leq \nu^*$, $\Delta RI = \emptyset$.

For $\nu > \nu^*$, the exact vectors are

$$\begin{aligned}
\text{II, } \textit{screening} &: (d, 0), \\
\text{II, } \textit{pooling} &: (0, 0), \\
\text{II, } \textit{exclusion} &: (k, -a_1), \\
\text{IX, } \textit{screening} &: (d, a_1), \\
\text{IX, } \textit{exclusion} &: (k, 0), \\
\text{XX, } \textit{exclusion} &: (d, 0).
\end{aligned}$$

On the residual segment, the exact attainable set is $\{\lambda(k, -a_1) : \lambda \in [0, 1]\}$ over the surviving proposal weights.

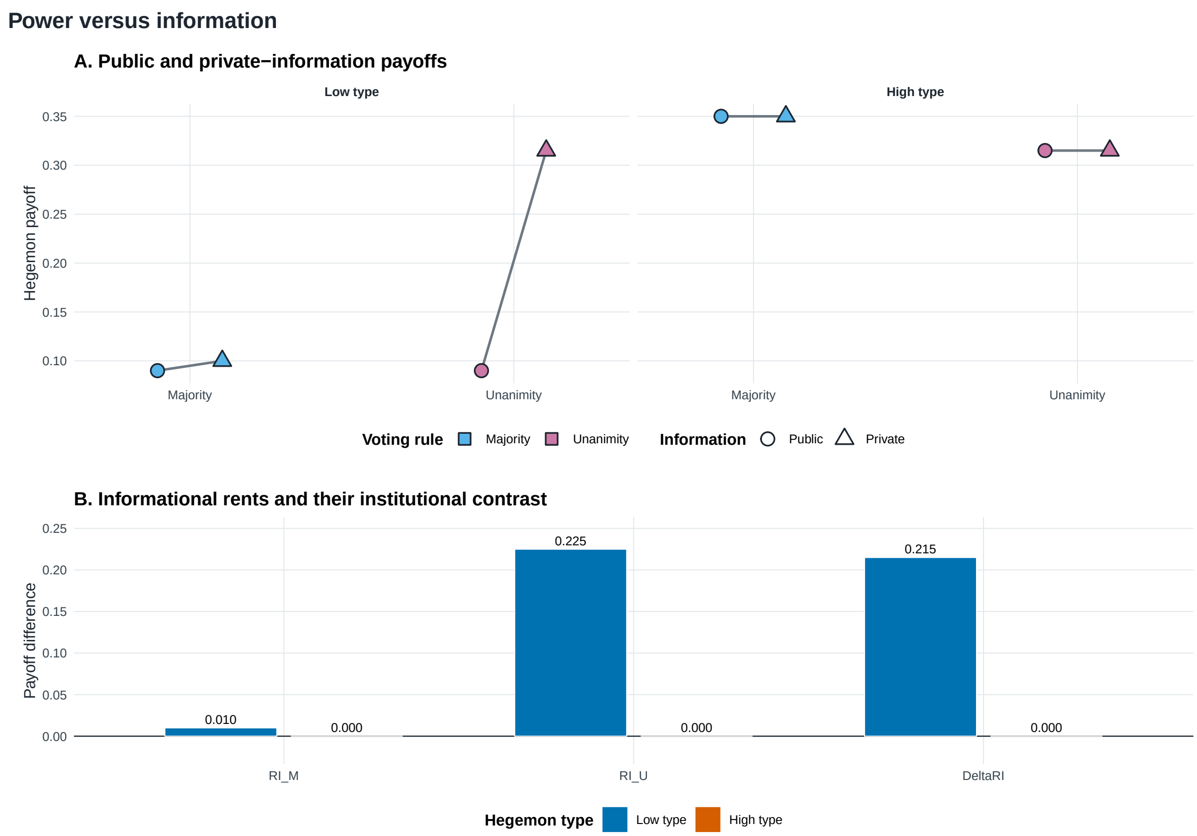
Table 5: Exact informational-rent correspondences and institutional contrast

Object	Cell	Exact set
RI_M	II, screening	$\{(0, 0)\}$
RI_M	II, pooling	$\{(d, 0)\}$
RI_M	II, exclusion	$\{(a_0, a_1)\}$
RI_M	II, exclusion–pooling segment	$\{\lambda(a_0, a_1) + (1 - \lambda)(d, 0)\}$
RI_M	IX, screening	$\{(0, -a_1)\}$
RI_M	IX, exclusion	$\{(a_0, 0)\}$
RI_M	XX, exclusion	$\{(0, 0)\}$
RI_U	$\nu = 0$	$\{(0, 0)\}$
RI_U	$0 < \nu \leq \nu^*$	$\emptyset$
RI_U	$\nu > \nu^*$	$\{(d, 0)\}$
ΔRI	$\nu = 0$, II	$\{(0, 0)\}$

Object	Cell	Exact set
ΔRI	$\nu = 0$, IX	$\{(0, a_1)\}$
ΔRI	$\nu = 0$, XX	$\{(0, 0)\}$
ΔRI	$0 < \nu \leq \nu^*$, every public region	$\emptyset$
ΔRI	$\nu > \nu^*$, II screening	$\{(d, 0)\}$
ΔRI	$\nu > \nu^*$, II pooling	$\{(0, 0)\}$
ΔRI	$\nu > \nu^*$, II exclusion	$\{(k, -a_1)\}$
ΔRI	$\nu > \nu^*$, II exclusion–pooling segment	$\{\lambda(k, -a_1) : \lambda \in [0, 1]\}$
ΔRI	$\nu > \nu^*$, IX screening	$\{(d, a_1)\}$
ΔRI	$\nu > \nu^*$, IX exclusion	$\{(k, 0)\}$
ΔRI	$\nu > \nu^*$, XX exclusion	$\{(d, 0)\}$

In every segment row, $\lambda \in [0, 1]$ is restricted to the proposal weights that survive the tie-break and binds all payoff and outcome components.

The sign pattern is transparent once each component is kept separate. Above the unanimity cutoff, the low type gains $d > 0$ when majority screens because unanimity pools and pays the high threshold. The same screening comparison is zero at $\nu = 0$, while the middle cell is empty. When majority excludes, the low type's gain is k, which is positive, zero, or negative according as βo_1 is above, equal to, or below o_0. The high type gains $a_1 > 0$ only when private majority screens even though public majority would exclude it; it loses a_1 when public majority includes it but private majority excludes. When both private rules pool, the informational-rent contrast is exactly zero.



Exact power−information decomposition at the working illustration (o0 = 0.10, o1 = 0.35, m = 4, beta = 0.90, nu = 0.80). Panel A compares each rule under public and private information. Panel B reports RI_M, RI_U, and DeltaRI = RI_U − RI_M component by component. At this point majority excludes and unanimity pools: the low−type values are 0.010, 0.225, and 0.215, while all high−type informational−rent values are zero. This is a working numerical illustration of the exact N7 formulas, not a calibration.

Figure 4: Power and information in the hegemon's payoff. The public benchmark captures the effect of making the hegemon's vote necessary when its type is known. Informational rent compares the private game with that benchmark, and ΔRI compares this rent across unanimity and majority without recombining types or selecting among multiple equilibria.

5 Discussion

5.1 Pivotality, substitutes, and contested strength

The model's central distinction is technological. Under majority, the proposer can replace the hegemon with another uninformed weak-state vote. Under unanimity, the hegemon's approval is an essential input. Removing the substitute changes both bargaining power and the way private information enters proposals.

The complete-information benchmark shows the power component. If the known hegemon is cheap enough, majority includes it and both rules pay the same discounted threshold. If it is expensive, majority excludes it while unanimity must still buy its vote. The informational comparison asks a different question: after netting out that public-type difference, how much does private information change each rule's payoff?

The answer is not simply that private information always benefits a hegemon. Pooling under unanimity benefits the low type because weak states price it as high. Screening under majority can remove that rent, while exclusion can make the comparison depend on whether the high threshold discounted to the present exceeds the low current disagreement payoff. The high type's rent changes only when its treatment differs across the private and public majority games.

The empty middle-belief cell is equally important for scope. It records the failure of every pure voting pattern under the declared belief and pivotality conditions. It does not supply a payoff for institutional comparison and does not justify filling the gap by interpolation.

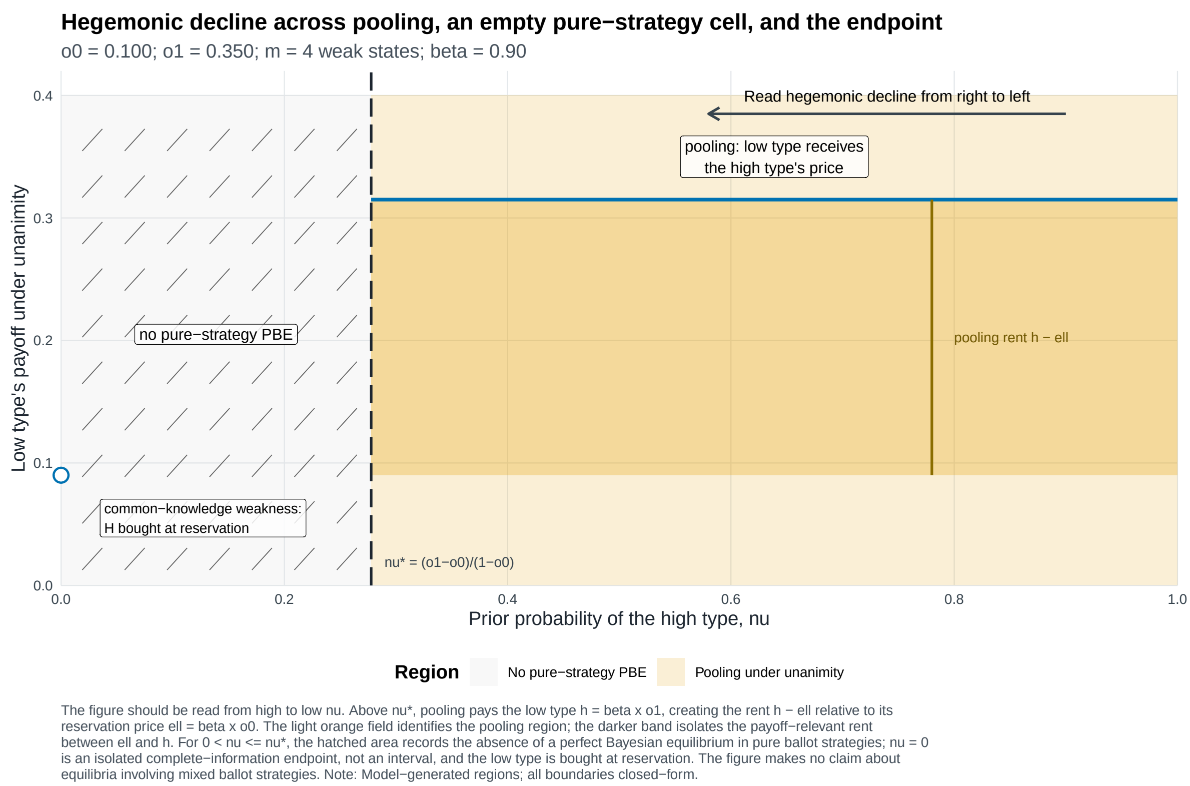


Figure 5: A schematic movement in beliefs about hegemonic strength. The hatched region is the set in which the unanimity game has no equilibrium in pure ballot strategies under the maintained concept. Historical annotations, if used, are illustrations rather than empirical tests or calibration.

5.2 OPEC, the WTO, and observable implications

In the OPEC interpretation, a proposed package can represent quota flexibility, an exception, or enforcement discretion. Saudi Arabia may know more than other members about the opportunity cost of accepting it. The model predicts that making its approval necessary removes a coalition substitute and can induce an offer calibrated to the higher possible reservation value. This is a mechanism illustration, not an estimate of OPEC bargaining.

For the WTO, consensus similarly makes a major state's vote necessary despite formally equal voting rights and the absence of exclusive proposal power. The model suggests looking for institutional concessions that arise because partners cannot confidently price the major state's willingness to accept. The relevant observable contrast is not merely whether the hegemon is included, but whether majority would provide a cheaper coalition of uninformed votes.

These implications differ from informational voting models in which individual votes aggregate dispersed signals (Feddersen and Pesendorfer 1998). Here weak states do not possess private signals about H's type. They respond to a public proposal and to the continuation induced by the publicly revealed vote vector. The nearby legislative-bargaining literature analyzes additional equilibrium concepts and strategic forms (Piazolo and Vanberg 2025; Glynia, Thum, and Xefteris 2026); our comparison deliberately remains within the pure ballot strategies stated above.

5.3 Limits

The analysis fixes the voting rule, keeps the institutional pie at one, gives proposal power only to weak states, and uses two types and two rounds. It does not analyze endogenous rule choice, agenda power for H, more than two types, sequential roll-call voting, or intrinsic agreement benefits. The restriction $m \geq 3$ excludes the three-player case. The strict discounting assumption excludes the corner $\beta = 1$, where additional indifferences would require a separate analysis.

The endpoint-support condition also matters. A type assigned zero probability by the prior cannot be resurrected after an off-path action. This makes the degenerate-prior cells coincide with their complete-information games. In the interior, both types remain in the prior's support and off-path actions by H can sustain a range of posterior beliefs.

6 Conclusion

Consensus can benefit a hegemon even when weak states control the agenda because unanimity removes substitutes for the hegemon's vote. Complete information reveals the familiar veto-power component. Private information adds a distinct component: when unanimity pools at the high threshold, a low type can earn rent; whether this exceeds its majority rent depends on the majority equilibrium and the outside-option region.

The decomposition disciplines the claim. The institutional informational-rent contrast is positive in some regions, zero when majority already pays the pooling price, sign-indeterminate through $k = \beta o_1 - o_0$ under exclusion, and empty where unanimity has no pure-strategy equilibrium. Formal equality of votes can therefore coexist with asymmetric informational power, but the effect is neither universal nor separable from the coalition alternatives created by the voting rule.

Appendix A: Protocol, beliefs, and incentives

A.1 Complete transition and payoff rules

For a proposal $s = (y, (x_j), r_i)$, let $a_H \in \{Y, N\}$ and let the weak responders' votes be a_{-i}. The proposer votes yes. If the quota is met, the proposal is implemented in full. Each weak respondent receives x_j, the proposer receives r_i, and H receives y after yes and $y + o_\theta$ after no. If the quota is not met in Round 1, the public vote vector leads to Round 2; every continuation payoff is then multiplied by β. If the quota is not met in Round 2, weak states receive zero and H receives o_θ.

The $y + o_\theta$ branch is necessary for a complete strategy specification. It is not an equilibrium description. Whenever majority passes without H, a proposer strictly prefers to move any positive y into its own residual, holding all votes fixed. Hence all equilibrium exclusions set $y = 0$ and pay H exactly o_θ.

A.2 Beliefs and ballot restrictions

Uninformed weak-state proposals and votes preserve the current belief. A vote by H updates by Bayes' rule when its denominator is positive. At a zero-denominator action in an interior-prior game, the posterior may range over $[0, 1]$, subject to structural consistency across histories that encode the same information. At $\nu = 0$ or $\nu = 1$, posterior support never expands.

A weak responder compares its allocation with the continuation that would follow if its vote switched passage into failure. It votes yes at equality. H maximizes its type-contingent continuation payoff and votes yes at exact indifference. These restrictions apply at every proposal, including off-path proposals.

Appendix B: Proofs

B.1 Proof of Proposition 4.1

In terminal majority, a weak responder's disagreement value is zero, so any nonnegative allocation induces yes under T^Y. Because $q \leq m$, the proposer can pass without H, set every payment to zero, and keep one. H is nonpivotal and strictly votes no because no yields $y + o$ rather than y. Moving any positive y to the proposer leaves passage unchanged, so equilibrium exclusion has $y = 0$ and H receives o.

In terminal unanimity, every weak responder accepts zero and H accepts exactly when $y \geq o$. The proposer therefore sets $y = o$.

Move to Round 1. The terminal continuation of a weak state before recognition under majority is $1/m$, hence $w = \beta/m$. The continuation of H is βo. Under unanimity, paying H exactly βo and each weak responder its continuation implements immediate agreement and leaves the proposer strictly more than delay because $\beta < 1$.

Under majority, inclusion costs $(q-2)w + \beta o$; exclusion costs $(q-1)w$. Inclusion is weakly cheaper exactly when $o \leq 1/m$. At equality the proposer is indifferent, and the proposal tie-break selects inclusion because H receives $\beta o < o$. This gives the stated payoffs. □

B.2 Proof of Proposition 4.2

The majority argument is the terminal-majority part of the preceding proof. Before recognition, symmetry gives each weak state expected payoff $1/m$.

Under unanimity, weak responders accept zero. Any accepted offer to H can be reduced to the lowest threshold consistent with its acceptance set. The only nonempty acceptance sets are the low type alone, implemented by $y = o_0$, and both types, implemented by $y = o_1$. The first gives the proposer $(1-\nu)(1-o_0)$; the second gives $1 - o_1$. The low offer is weakly preferred exactly when

$$\nu \leq \frac{o_1 - o_0}{1 - o_0} = \nu^*.$$

At equality the low offer gives H the lower expected payoff and is selected. □

B.3 Proof of Proposition 4.3

A responding weak state votes yes exactly when $x_j \geq w = \beta/m$. Let k be the number of such responders. If $k \geq q-1$, the weak votes pass the proposal without H; both types of H vote no and receive $y + o_\theta$. If $k = q - 2$, H is pivotal and type θ votes yes exactly when $y \geq t_\theta = \beta o_\theta$. If $k \leq q - 3$, the proposal fails regardless of H's vote.

Within each acceptance class, reducing weak payments to w, reducing y to the relevant type threshold, and assigning the remaining pie to the proposer weakly increases its payoff. This leaves four candidates: exclusion with payoff E, screening with payoff $S(\nu)$, pooling with payoff P, and deliberate delay with payoff w. Exclusion is always feasible and

$$E - w = 1 - \frac{\beta q}{m} > 0,$$

so delay is never selected. A screening or pooling candidate that could beat exclusion is automatically feasible.

The decisive payoff differences are

$$P - E = \beta(1/m - o_1)$$

and

$$S(\nu) - E = (1 - \nu)\beta(1/m - o_0) - \nu(1 - \beta q/m).$$

Where pooling can beat exclusion, comparing screening with pooling yields ν_{SP}; comparing screening with exclusion yields ν_{SE}. Substituting the signs of $o_0 - 1/m$ and $o_1 - 1/m$ produces the five cases in Proposition 4.3.

At screening ties, its expected payoff to H is strictly smaller than that of the tied alternative, so the proposal tie-break selects screening. When $o_1 = 1/m$, exclusion and pooling give the proposer the same payoff. Their payoffs to H determine the stated selection; exact equality leaves their entire proposal segment. All selected proposals exhaust the pie. Finally, permuting the identities of weak responders does not change any comparison, which establishes the reported identity multiplicity. □

B.4 Proof of Proposition 4.4

Transport the terminal unanimity continuation into Round-1 units. If the posterior is η, a weak state's continuation is

$$W(\eta) = \begin{cases} (1-\eta)A, & \eta \le \nu^*, \\ B, & \eta > \nu^*. \end{cases}$$

Thus $W(\eta) \in [B, A]$, and $x_j = A$ forces every weak responder to vote yes under any admissible posterior.

At $\nu = 0$, support preservation fixes every posterior at zero. The proposal

$$y = \ell, \qquad x_j = A, \qquad r_i = 1 - \ell - (m-1)A = A + 1 - \beta$$

is accepted. Each responder and H is exactly indifferent and votes yes. Any lower payment loses a required vote; any higher payment reduces the proposer's residual. A delay gives the proposer A, so immediate agreement is strictly better by $1 - \beta$.

For $\nu > \nu^*$, the analogous pooling proposal is

$$y = h, \qquad x_j = B, \qquad r_i = 1 - h - (m-1)B = B + 1 - \beta.$$

Both types of H and every weak responder accept, and immediate agreement again exceeds delay by $1 - \beta$.

For completeness, write $u = \min_j x_j$ after an arbitrary proposal and order the strategy of H as low-type/high-type. At $\nu = 0$, posterior support is always the low type. The profile (Y, Y) is sequentially rational if $u < A$, or if $u \ge A$ and $y \ge h$; (N, N) is rational if $u \ge A$ and $y < \ell$; and (Y, N) is rational if $u \ge A$ and $\ell \le y < h$. The profile (N, Y) is never rational. These cases are mutually exclusive and exhaustive.

For $\nu^* < \nu < 1$, the profile (Y, Y) is rational if $u < B$, or if $u \ge B$ and $y \ge h$; (N, N) is rational if $u \ge B$ and $y < h$; and neither separating profile is rational. Bayes gives the current prior after an on-profile pooled action. When yes is a zero-probability action, an admissible posterior η_Y supports all weak yes votes exactly when $W(\eta_Y) \le u$, which is possible in the stated no profile because

$u \geq B$. At $\nu = 1$, support preservation fixes the posterior at one and the same two pooled-profile conditions apply. Thus every proposal in both existence domains has a pure, sequentially rational response completion.

It remains to show the empty cell. For $0 < \nu \leq \nu^*$, consider

$$s^\dagger: \quad y = \ell, \quad x_j = A \text{ for every weak responder}, \quad r_i = 1 - \ell - (m-1)A.$$

All weak responders must vote yes. Enumerate the four pure type-contingent profiles for H, ordered low/high:

1. Under (Y, Y), the high type receives $\ell < h$ and profitably votes no to obtain continuation h.
2. Under (N, N), the low type's no gives ℓ, so T^Y requires yes at equality.
3. Under (Y, N), the low type can imitate the high type's no. That action reveals the high type and yields continuation $h > \ell$.
4. Under (N, Y), the high type can imitate the low type's no and obtain continuation h, while the prescribed comparison cannot support strict no against yes at the offer.

No pure profile is sequentially rational after $s^\dagger$. Because a PBE must specify a valid pure response after every feasible proposal, no pure-ballot PBE exists in this cell. This also explains the discontinuity: at $\nu = 0$, support preservation rules out the posterior that drives the imitation argument; for every positive ν, both types are possible. □

B.5 Proof of Proposition 4.5

The private majority payoff vectors for H are

$$V_M^S = (\beta o_0, \beta o_1), \quad V_M^P = (\beta o_1, \beta o_1), \quad V_M^E = (o_0, o_1).$$

The private unanimity vectors are $(\beta o_0, \beta o_1)$ at $\nu = 0$, empty for $0 < \nu \leq \nu^*$, and $(\beta o_1, \beta o_1)$ for $\nu > \nu^*$. Componentwise subtraction gives the proposition. On any proposal segment, subtraction is affine in the common proposal weight, so the payoff set and outcome set remain atomic. □

B.6 Proofs of Propositions 4.6 and 4.7

From Proposition 4.1, the public payoff vectors are

$$V_M^{pub} = (p_M(o_0), p_M(o_1)), \qquad V_U^{pub} = (\beta o_0, \beta o_1).$$

Subtracting these from each private vector gives the table in Proposition 4.6. In particular, unanimity is payoff-equivalent to its public endpoint at $\nu = 0$, is empty in the middle cell, and pools above the cutoff, which yields $(d, 0)$.

Subtract RI_M from RI_U. In the high-belief cell, screening gives $(d, 0)$ in region II and (d, a_1) in region IX; pooling gives zero; exclusion gives $(k, -a_1)$, $(k, 0)$, or $(d, 0)$ in regions II, IX, or XX. The endpoint follows by the same calculation. An empty source set remains empty. For the residual segment, affine subtraction with its common weight λ yields the one-dimensional segment stated in the propositions, not the Cartesian product of marginal envelopes. $\square$

Appendix C: Endpoints, exact sets, and illustration

C.1 Endpoint equivalence

At $\nu = 0$, posterior support is the singleton low type. The private unanimity proposal and payoff coincide with the public low-type game. At $\nu = 1$, posterior support is the singleton high type, and the private pooling proposal and payoff coincide with the public high-type game. The same equivalence holds under majority: applying the relevant private class at each endpoint reproduces public inclusion or exclusion, including the $o_\theta = 1/m$ tie-break. Thus the endpoints are literal complete-information games, not one-sided limits.

C.2 Exact correspondence and envelopes

Most parameter cells are singleton payoff vectors. At $o_1 = 1/m$, exact indifference between exclusion and pooling can leave a proposal segment. If λ is the exclusion weight, every reported object uses that same λ. The lower and upper envelope of any component may summarize the segment, but coordinated marginal envelopes do not imply that all pairwise combinations are attainable.

For majority-rule informational rents, the exact componentwise envelopes are

$$RI_M(0) \in [\min\{a_0, d\}, \max\{a_0, d\}], \qquad RI_M(1) \in [0, a_1].$$

For both the private institutional payoff contrast and ΔRI, they are

$$\text{low type} \in [\min\{0, k\}, \max\{0, k\}], \qquad \text{high type} \in [-a_1, 0].$$

In each case these are the marginal envelopes of a single line segment. The same λ indexes both coordinates, so the Cartesian product of the two intervals is not attainable.

Where $0 < \nu \leq \nu^*$, the private unanimity correspondence is empty. Accordingly, V_U^{priv}, RI_U, and ΔRI are empty. No payoff, sign, institutional ranking, or interpolation is assigned to that cell.

C.3 Worked values

Return to $m = 4$, $q = 3$, $\beta = 0.9$, $o_0 = 0.10$, and $o_1 = 0.35$. Besides $\nu^* = 0.2778$, the majority screening–exclusion cutoff is

$$\nu_{SE} = \frac{0.9(0.25 - 0.10)}{0.9(0.25 - 0.10) + 1 - 0.9(3/4)} = 0.2935.$$

At $\nu = 0.80$, majority excludes and unanimity pools. The public payoff vectors are

$$V_M^{pub} = (0.09, 0.35), \qquad V_U^{pub} = (0.09, 0.315),$$

while the private payoff vectors are

$$V_M^{priv} = (0.10, 0.35), \qquad V_U^{priv} = (0.315, 0.315).$$

Therefore

$$RI_M = (0.01, 0), \qquad RI_U = (0.225, 0), \qquad \Delta RI = (0.215, 0).$$

These numbers illustrate one point in region IX; they do not average over parameter values or estimate a real institution.

Appendix D: Notation

Table 6: Notation for the essential-input bargaining game

Symbol	Meaning
H	Privately informed hegemon
W, m	Set and number of weak states; $m \geq 3$
$N = m + 1$	Total number of states
q	Majority quota, $\lfloor N/2 \rfloor + 1$
θ	Hegemon type, low 0 or high 1
ν	Prior probability of the high type
o_θ	Type-θ terminal disagreement payoff
β	One-round discount factor, strictly between zero and one
y	Institutional concession assigned to H
$\bar{y}$	Maximum feasible institutional concession; $o_1 \leq \bar{y} \leq 1$
x_j	Allocation assigned to weak responder j
r_i	Residual assigned to weak proposer i
$w = \beta/m$	First-round continuation value of a weak state under majority
$t_\theta = \beta o_\theta$	First-round threshold of type θ when pivotal
ν^*	Terminal unanimity cutoff, $(o_1 - o_0)/(1 - o_0)$
a_θ	Timing wedge, $(1 - \beta)o_\theta$
d	Discounted type gap, $\beta(o_1 - o_0)$
k	Cross-date quantity, $\beta o_1 - o_0$
V_g^{priv}, V_g^{pub}	Private- and public-information payoff correspondences under rule g
RI_g	Informational-rent correspondence $V_g^{priv} - V_g^{pub}$
ΔRI	Unanimity-minus-majority informational-rent contrast

References

Baron, David P., and John A. Ferejohn. 1989. “Bargaining in Legislatures.” *American Political Science Review* 83 (4): 1181–1206.

Eraslan, Hülya, and Kirill S. Evdokimov. 2019. “Legislative and Multilateral Bargaining.” *Annual Review of Economics* 11: 443–72.

Fattouh, Bassam, and Lavan Mahadeva. 2013. “OPEC: What Difference Has It Made?” *Annual Review of Resource Economics* 5: 427–43.

Feddersen, Timothy J., and Wolfgang Pesendorfer. 1998. “Convicting the Innocent: The Inferiority of Unanimous Jury Verdicts Under Strategic Voting.” *American Political Science Review* 92 (1): 23–35. https://doi.org/10.2307/2585926.

Fudenberg, Drew, and Jean Tirole. 1991. *Game Theory*. Cambridge, MA: MIT Press.

Glynia, Nektaria, Marcel Thum, and Dimitrios Xefteris. 2026. “Unanimity Versus Majority: Proposing Under Incomplete Information.” *Public Choice*. https://doi.org/10.1007/s11127-026-01383-9.

Koremenos, Barbara, Charles Lipson, and Duncan Snidal. 2001. “The Rational Design of International Institutions.” *International Organization* 55 (4): 761–99.

Kreps, David M., and Robert Wilson. 1982. “Sequential Equilibria.” *Econometrica* 50 (4): 863–94.

Miller, Luis, Maria Montero, and Christoph Vanberg. 2018. “Legislative Bargaining with Heterogeneous Disagreement Values: Theory and Experiments.” *Games and Economic Behavior* 107: 60–92. https://doi.org/10.1016/j.geb.2017.11.003.

Nakov, Anton, and Galo Nuño. 2013. “Saudi Arabia and the Oil Market.” *The Economic Journal* 123 (573): 1333–62.

Osborne, Martin J., and Ariel Rubinstein. 1990. *Bargaining and Markets*. San Diego: Academic Press.

Piazolo, David, and Christoph Vanberg. 2025. “Legislative Bargaining with Private Information: A Comparison of Majority and Unanimity Rule.” *Games and Economic Behavior* 153: 499–522. https://doi.org/10.1016/j.geb.2025.07.010.

Steinberg, Richard H. 2002. “In the Shadow of Law or Power? Consensus-Based Bargaining and Outcomes in the GATT/WTO.” *International Organization* 56 (2): 339–74.

Stone, Randall W. 2011. *Controlling Institutions: International Organizations and the Global Economy*. Cambridge: Cambridge University Press.

Winter, Eyal. 1996. "Voting and Vetoing." *American Political Science Review* 90 (4): 813–23. https://doi.org/10.2307/2945844.